

The GNS representation of the target trace extension

MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation

def

The chosen state on the existing target supplies its cyclic GNS representation.

Statement

Let A be the completed CAR algebra and τ its normalized trace. Fix a representative shell family and its chosen unitary links L_j in the selected atomic representation π . Write $T = C^*(\pi(A), \{L_j\})$ for the resulting concrete target and $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, for its faithful unital source map. This fixes one target and one source map throughout. Let ϕ be the chosen state extension of τ to T . Define $\rho_\phi : T \rightarrow B(H_\phi)$ to be its GNS representation.

Definition

Choose ϕ from `exists_state_extension_trace` and regard that same functional as a positive linear map. Form the GNS completion of T modulo its null vectors, set H_ϕ to that Hilbert space, and let $\xi_\phi = [1]$. The induced left action is

$$\rho_\phi(t)[b] = [tb], \quad \phi(t) = \langle \xi_\phi, \rho_\phi(t)\xi_\phi \rangle.$$

The right-hand side of the definition is exactly the GNS star homomorphism of `traceExtensionPositive` family. This positive-map bundle has the same values as ϕ ; it is not another state. Normalization gives $\|\xi_\phi\| = 1$, so $\xi_\phi \neq 0$ and H_ϕ is nonzero. The GNS construction gives the dense T -orbit. Its restriction realizes τ because $\phi \circ \iota = \tau$. The actual-representation source-cyclicity theorem applies to these two facts and proves the source orbit dense; it is not an assumption smuggled into the definition.

Assumptions

A representative shell family is fixed first. The chosen extension is positive and normalized; its traciality is not needed to perform the GNS construction.

Conclusion

With its canonical unit vector ξ_ϕ , this representation realizes ϕ and has dense target orbit. The separately proved source-cyclicity result also gives

$$\overline{\{\rho_\phi(\iota(a))\xi_\phi : a \in A\}} = H_\phi.$$

Consequently H_ϕ is separable, since A is separable and the displayed orbit map is continuous.

Separability here is a conclusion about the Hilbert space. The algebra T is not assumed norm separable. Injectivity of ρ_ϕ is a separate theorem; irreducibility or purity of ϕ is not asserted.

Main citations

- Definition and its exact construction — `MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation`
- The fixed concrete target and source map — `MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`
- Faithfulness of the source embedding — `MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom_injective`
- Existence used to choose the state — `MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace`
- The chosen extension — `MathlibAnnex.CStarAlgebra.CAR.traceExtension`
- The positive-map bundle — `MathlibAnnex.CStarAlgebra.CAR.traceExtensionPositive`
- The target-state GNS Hilbert space — `MathlibAnnex.CStarAlgebra.CAR.TracialHilbertSpace`
- The canonical target-state vector — `MathlibAnnex.CStarAlgebra.CAR.tracialVector`
- Normalization of the vector — `MathlibAnnex.CStarAlgebra.CAR.norm_tracialVector`
- Nonzero target-state vector — `MathlibAnnex.CStarAlgebra.CAR.tracialVector_ne_zero`
- Nontriviality of its Hilbert space — `MathlibAnnex.CStarAlgebra.CAR.nontrivial_tracialHilbertSpace`
- The GNS vector-state identity — `MathlibAnnex.CStarAlgebra.CAR.inner_tracialVector_tracialRepresentation`
- The source restriction realizes the trace — `MathlibAnnex.CStarAlgebra.CAR.vectorFunctional_tracialRepresentation_source`
- Target cyclicity from GNS — `MathlibAnnex.CStarAlgebra.CAR.denseRange_tracialRepresentation_orbit`
- Source cyclicity is proved — `MathlibAnnex.CStarAlgebra.CAR.denseRange_tracialRepresentation_source_orbit`
- Separable target-state GNS space — `MathlibAnnex.CStarAlgebra.CAR.separableSpace_tracialHilbertSpace`

Supporting route explanation

Lean source signature (exact)

```
noncomputable def tracialRepresentation (family : RepresentativeShellFamily) :
  Representation (ShellFamilyTarget family) (TracialHilbertSpace family) :=
  (traceExtensionPositive family).gnsStarAlgHom
```

In the source	Mathematical meaning
<code>family : RepresentativeShellFamily;</code> <code>ShellFamilyTarget family</code>	The supplied CAR shell family fixes the same target $T = C^*(\pi(A), \{L_j\})$ and source $\iota : A \rightarrow T$.
<code>traceExtensionPositive family</code>	The positive-linear-map bundle of the chosen state extension $\phi : T \rightarrow \mathbb{C}$ of the CAR trace. Its values are exactly $\phi(t)$; it is not a second state.

In the source	Mathematical meaning
TracialHilbertSpace family	The complete GNS Hilbert space H_ϕ obtained by completing T modulo the null vectors of ϕ , with canonical unit vector $\xi_\phi = [1]$.
Representation (ShellFamilyTarget family) (TracialHilbertSpace family)	The output is a unital star representation $\rho_\phi : T \rightarrow B(H_\phi)$.
(traceExtensionPositive family).gnsStarAlgHom	The full RHS is the GNS left action $\rho_\phi(t)[b] = [tb]$. Its vector-functional identity, unit-vector norm, density, source cyclicity and separability are separately proved properties of this action, not fields omitted from the definition. Irreducibility is not asserted.

Further source notes: The norm, vector-functional, density and separability statements are separately cited theorems, not omitted fields of this definition.

Earlier published Card and PDF

▼ Exact content identity

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